



INDUSTRIAL CONNECTIVITY / ORDER-BASED SOLUTION STUDY

Connecting legacy interfaces. Enabling practical integration.

USB, Ethernet, RS-232 / RS-485 and 4G connectivity components for a controlled engineering and validation approach.

6 ITEMS

ONE EACH IN THE PROVISIONAL ORDER

Buyer: Expede-Tech Research & Development Pvt. Ltd. | Nayka, Kheda, Gujarat

STPO-2425-401 | 22 November 2024 | Provisional order; final award, installation and results are not established.

01 / MATERIAL-TO-SOLUTION MAPPING

Six components. Distinct connectivity roles.

Source: provisional order STPO-2425-401. The technical roles below require model and interface verification.

Ordered item / quantity	Intended role and selection check
JioFi 4G dongle / 1	Candidate mobile internet access for an approved host. Confirm actual model, host connection method, SIM / plan and permitted use.
Ethernet-to-USB converter / 1	Candidate wired-network interface for a USB host. Confirm direction, OS driver, USB type and supported Ethernet speed.
RS-485-to-RS-232 converter / 1	Candidate serial electrical-interface conversion. Confirm active conversion, power, isolation, wiring mode and device compatibility.
USB-to-RS232/485 converter / 1	Candidate host serial interface. Confirm selected serial mode, chipset, driver and supported communication settings.
USB-to-serial DB9 adapter / 1	Candidate USB-to-serial host connection. Confirm RS-232 electrical level, DB9 pinout and DTE / DCE cable requirement.
RJ45 male to RS232 DB9 female / 1	Description is ambiguous: verify active Ethernet-to-serial gateway versus a passive serial / console adapter. Connector shape does not establish function.

The order requests supply. Implementation design and validation steps in this study are proposed engineering practice, not a completion report.

02 / FUNCTIONAL SOLUTION SCHEMATIC

Choose the correct path for each interface

Alternative connection routes are shown separately; the six items are not intended to form one mandatory series chain.

ENGINEERING HOST

Customer PC / test application
Host and software not in this order

→

USB SERIAL OPTION

USB-to-RS232/485 OR
USB-to-serial DB9 adapter

→

SERIAL DEVICE

Verified RS-232 or RS-485
Application protocol unchanged

RS-232 SOURCE

Device / host with a verified
RS-232 electrical interface

→

ACTIVE CONVERTER

RS-232 to RS-485 route
Power and pinout to be checked

→

RS-485 DEVICE

Verify 2-wire / 4-wire mode
and OEM wiring requirements

NETWORK OPTIONS: USB host → verified USB/Ethernet adapter → approved LAN. Approved host → verified JioFi connection → 4G internet.

RJ45/DB9 item remains outside the network path until its datasheet confirms its function. No direct internet-to-machine connection is proposed.

03 / MICRO-LEVEL INTEGRATION PLAN

Validate the interface before connecting the process

Proposed commissioning checklist. Exact models, machine endpoints and application protocols are absent from the order.

Engineering gate	Checks and acceptance evidence
01 / Identify interfaces	Record device model, port type, electrical level, connector pinout, cable and host OS. Distinguish RS-232, RS-485 and Ethernet explicitly.
02 / Prepare the host	Use the approved driver; record adapter identity, COM port / network-interface name and version. Capture the baseline before changing settings.
03 / Set communication	Match baud rate, data bits, parity, stop bits and flow control to the endpoint. Verify protocol / address settings with the equipment owner.
04 / Verify wiring	Check Tx/Rx and ground for RS-232; confirm RS-485 polarity and wiring mode from both OEM manuals. Assess isolation, shielding and termination requirements.
05 / Bench and application test	Use an approved loopback or test endpoint where supported; then validate a representative application exchange. Check receive / transmit errors and reconnection.
06 / Controlled handover	Document wiring, driver and settings; label adapters; retain test results and rollback steps. Obtain customer acceptance before production use.

A serial converter changes the electrical interface; protocol translation or machine integration needs separately verified application capability.

04 / DELIVERY AND SUPPORT WORKFLOW

From requirement capture to verified connectivity

A proposed service approach for the component supply; no AMC term, staffing or numerical SLA is stated in the source order.

01 DISCOVER

Identify the host, machine interface, protocol, environment and intended business use.

02 QUALIFY

Match product datasheets, power, drivers, pinout and active / passive function.

03 SUPPLY AND CHECK

Verify ordered quantities, packaging, model / serial records and compatibility before deployment.

04 CONFIGURE AND TEST

Use an approved change window; test the selected connection path and application exchange.

05 ACCEPT AND HAND OVER

Record settings, wiring and test evidence. Confirm customer acceptance and support ownership.

06 SUPPORT AND REVIEW

Log symptoms and connection state; isolate host, adapter, cable, endpoint or application faults.

05 / VALUE AND PROJECT BASIS

Practical integration starts with correct selection

D-TECH capability focus: translating a component requirement into an understandable, testable connection design.

COMPATIBILITY

Identify the electrical interface, host driver and pinout before selecting the cable or adapter.

CONTROLLED VALIDATION

Bench checks and application tests provide a structured route to customer acceptance.

SUPPORT READINESS

Documented wiring and settings help isolate future faults and support replacement decisions.

Source and design basis

Buyer: Expede-Tech Research & Development Private Limited, Nayka, Kheda. Source: one-page provisional order STPO-2425-401 dated 22 November 2024. Its corporate address and website identify Jindal Textiles; supplied logo / photos provide brand context.

This is an order-based solution study. Delivery, installation, operating outcomes and final award are not verified. Product models, endpoint topology and protocols require validation. Commercial figures are omitted.

Engineering reference: Moxa NPort IA5000A-I/O manual - distinct Ethernet and serial port definitions / pinouts. Reference only; Moxa is not specified in this order.

D-TECH SUPPORT | support.dtechindia.com **+91 99989 03042**